

Can Deficits Finance Themselves?

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Based on Angeletos, Lian, Wolf (Econometrica, 2024)

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Motivation

- ▶ Standard view: deficits require future taxes
- ▶ Based on Ricardian equivalence
- ▶ But in reality:
 - ▶ High MPC households
 - ▶ Nominal rigidities
 - ▶ Active monetary policy

Question: Can deficits reduce their own fiscal cost?

Preview of Answer

- ▶ Deficits can be partially self-financing
- ▶ Two channels:
 - ▶ Output $\uparrow \Rightarrow$ tax revenue $\uparrow$
 - ▶ Inflation $\uparrow \Rightarrow$ real debt $\downarrow$

Key idea: Fiscal policy affects equilibrium outcomes that feed back into debt.

Framework

- ▶ New Keynesian model
- ▶ Heterogeneous agents (high MPC)
- ▶ Sticky prices
- ▶ Monetary policy: interest rate rule

Non-Ricardian environment: households do not fully offset fiscal policy

Key Intuition

1. Deficit increases disposable income
2. High MPC households consume more
3. Output rises
4. Inflation rises
5. Real value of debt falls

Government Budget Constraint

$$B_t = (1 + r_{t-1})B_{t-1} + G_t - T_t$$

Divide by GDP:

$$\frac{B_t}{Y_t} = \frac{1 + r_{t-1}}{1 + g_t} \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t}$$

- ▶ Standard view: deficits increase debt
- ▶ Paper: Y_t and inflation are endogenous

Channel 1: Output Effect

- ▶ Fiscal expansion increases demand
- ▶ With high MPC:

$$\Delta Y_t > 0$$

- ▶ Tax revenue increases:

$$\Delta T_t = \tau \Delta Y_t$$

Result: smaller net fiscal cost

Channel 2: Inflation Effect

Fisher equation:

$$r_t \approx i_t - \pi_t$$

- ▶ Higher inflation π_t
- ▶ Reduces real value of nominal debt:

$$\frac{B_t}{P_t} \downarrow$$

Interpretation: inflation acts like a tax on bondholders

Decomposition of Deficit Financing

$$\underbrace{\varepsilon_0}_{\text{deficit}} = \tau_d \underbrace{\left(\varepsilon_0 + \sum_{k=0}^{\infty} \beta^k \mathbb{E}_0[d_k] \right)}_{\text{fiscal adjustment}} + \tau_y \underbrace{\left(\sum_{k=0}^{\infty} \beta^k \mathbb{E}_0[y_k] \right)}_{\text{self-financing via tax base}} + \underbrace{\frac{D^{ss}}{Y^{ss}} (\pi_0 - \mathbb{E}_{-1}[\pi_0])}_{\text{self-financing via debt erosion}}$$

- ▶ τ_d : fiscal adjustment (future repayment)
- ▶ τ_y : tax rate applied to output

Figure 1

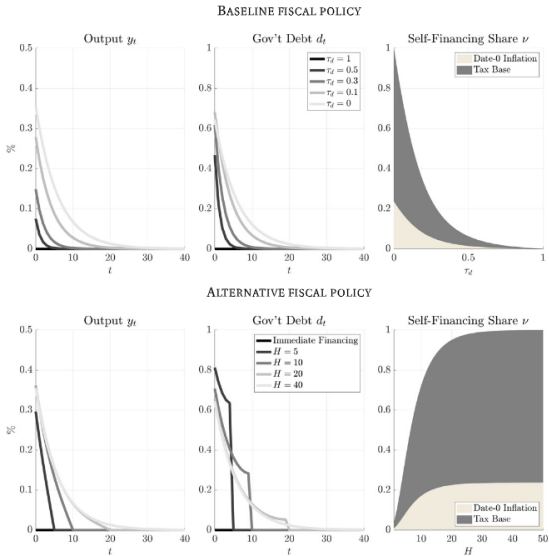


FIGURE 1.—Top panel: Impulse responses of output y_t , government debt d_t , and the self-financing share ν to a shock ε_0 equal to 1% of steady-state output, as a function of τ_d . Bottom panel: Same as above, but as a function of H .

Main Theoretical Result

Under:

- ▶ Delayed fiscal adjustment
- ▶ Passive monetary policy

In limit debt returns to baseline without future surpluses

Full self-financing is possible (in theory)

Policy Implications

- ▶ Fiscal policy more powerful
- ▶ Austerity may backfire
- ▶ Debt sustainability depends on:
 - ▶ Monetary policy
 - ▶ Inflation dynamics

Limitations

- ▶ Depends on:
 - ▶ MPC estimates
 - ▶ Price rigidity
- ▶ Inflation costs not modeled in detail
- ▶ Empirical magnitude uncertain

Conclusion

- ▶ Deficits are not free
- ▶ But can partially finance themselves

Main idea: Debt sustainability is a general equilibrium outcome